

ANILA IJAZ

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PROFESSIONAL EXPERIENCE

AI Research (Student Assistant) | TU Berlin (IWF), Germany

03/2026 – Present

UltraFaser4EDM Project

- Building **ML pipelines** for **time series anomaly detection** on signals, cutting false positives by 20%
- Developing a **SHAP-based explainability (XAI)** layer that makes AI model decisions readable
- Structuring **MLOps** practices for experiment tracking and versioning via **GitHub Actions**
- Translating domain expert requirements into working software through iterative collaboration
- Documenting model iterations and pipeline decisions, cutting new-member onboarding time

AI Research (Student Assistant) | TU Berlin (SNET Labs), Germany

05/2025 – 12/2025

WERK 4.0 Project

- Built an **LLM-based AI pipeline** from proof-of-concept to working prototype, cutting manual effort by 30%
- Integrated **REST APIs** for real-time **NLP** data ingestion, lifting accuracy by 25%
- Maintained **reproducible pipelines** with structured documentation to support long-term code quality
- Followed **agile/scrum** iteration cycles with regular reviews in a cross-functional team of 5–10 people
- Presented complex AI solutions clearly to both technical and non-technical stakeholders in workshop-style reviews

Software Engineer – Backend | NetSol Technologies, Pakistan

08/2022 – 09/2024

CMS Project

- Built and maintained backend services for enterprise financial clients, focusing on clean architecture, data integrity, and maintainability
- Optimised **SQL**-based reporting pipelines, improving data access efficiency by 40% through query refactoring and structural improvements
- Implemented data transformation logic to support downstream analytics and reporting systems
- Participated in code reviews and cross-team integration work, contributing to shared quality standards across the engineering organisation
- Aligned technical implementation with client requirements alongside product owners

Software Developer | Studio Mega Fury, Pakistan

06/2021 – 07/2022

- Developed modular backend components in **C#** and **C++** using clean code principles for scalability
- Participated in iterative development cycles including code reviews, debugging, and documentation
- Delivered tasks independently within a small, self-organised agile team

SELECTED PROJECTS

Fleet Document Intelligence Platform [[GitHub](#)]

Python, LangChain, LangGraph, LLMs, RAG, pgvector,

FastAPI, Docker, Render, CI/CD

- **Problem:** Fleet leasing teams spent hours manually extracting data from invoices, contracts, and damage reports
- **Solution:** Built end-to-end **LangGraph agentic triage pipeline** with pgvector RAG chatbot, deployed on **Render**
- **Impact:** Reduced manual document processing by **60%**; 16-test CI/CD suite ensured zero-regression production deploys

InsightRAG – Production-Grade RAG Pipeline [\[GitHub\]](#) *Python, LangChain, FAISS, BM25, LLMs, FastAPI, Docker*

- **Problem:** Naive RAG systems miss relevant context and are vulnerable to prompt-injection, limiting reliable enterprise Q&A over large document sets
- **Solution:** Built a production-grade **RAG pipeline** with hybrid retrieval (BGE + BM25, RRF fusion), a fine-tuned cross-encoder reranker, and prompt-injection guardrails
- **Impact:** Improved retrieval relevance and answer quality, validated end-to-end with the **RAGAS** framework; containerised with Docker and CI/CD for reproducible deployment

Intelligent Healthcare Chatbot [\[GitHub\]](#) *Python, LLMs, RAG, FastAPI*

- **Problem:** Patients struggle to get reliable, accessible answers to medical queries without combing through dense clinical sources
- **Solution:** Built a modular **RAG-based AI assistant** with clean separation across retrieval, generation, and serving layers, served via **FastAPI**
- **Impact:** Delivered grounded, source-backed responses with structured logging and evaluation pipelines, making the system testable, extensible, and production-ready

Real-Time Object & Shape Detection System [\[GitHub\]](#) *Python, OpenCV, YOLOv8, Deep Learning*

- **Problem:** No unified pipeline for deep-learning object detection and shape analysis in real time
- **Solution:** Integrated **YOLOv8** with OpenCV contour analysis into a single modular pipeline
- **Impact:** Achieved real-time inference at **23+ FPS**; reduced false detection rate by **35%** vs. baseline model

[Explore more projects on Github](#)

TECHNICAL SKILLS

Languages:	Python, SQL, C#, JavaScript, TypeScript, C++
AI & LLMs:	LLMs, RAG Pipelines, LangChain, LangGraph, Agentic AI, NLP, Vector Databases (pgvector), Embedding Models, Prompt Engineering, MLOps, MLflow
ML & Data:	PyTorch, TensorFlow, Scikit-learn, Time Series Analysis, Anomaly Detection, Computer Vision (YOLOv8, OpenCV), Explainable AI (SHAP)
Cloud & Infra.:	Docker, Render, CI/CD, GitHub Actions, Linux/Bash, Jupyter
Engineering:	Clean Code, Modular Architecture, Design Patterns, Code Review, Unit Testing, Agile/Scrum, REST APIs, FastAPI, Flask
Human Languages:	English (C1), German (A2), Urdu (Native)

EDUCATION

MSc Artificial Intelligence Brandenburg University of Technology (BTU), Germany	10/2024 – 08/2026
BS Computer Science University of Engineering & Technology (UET) Lahore, Pakistan	2017 – 2021

CERTIFICATIONS

Google Generative AI LLMs, Prompt Engineering, AI Applications – Badge Awarded	2024
Image Processing Workshop Certificate of Excellence – Computer Vision	2019

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

- **Coordinator, ProSoft Society** – Organised tech events, workshops, and programming competitions for the CS department at UET Lahore
- **Head Member, JZT Thalassemia Society** – Led awareness campaigns, blood donation drives, and community outreach initiatives